

Design a Rate Limiter - Step 4: Wrap Up

1. Algorithms Covered

The following rate limiting algorithms were discussed:

- Token Bucket
- Leaking Bucket
- Fixed Window
- Sliding Window Log
- Sliding Window Counter

2. Topics Covered in Design

- System architecture
- Distributed rate limiting
- Performance optimization
- Monitoring

3. Additional Talking Points (If Time Allows)

Hard vs Soft Rate Limiting

- ✓ Hard Limit

None

Requests cannot exceed the threshold

✓ **Soft Limit**

None

Requests can exceed threshold for a short period

Rate Limiting at Different Layers

- Chapter focused on:
 - **Application Layer (Layer 7 - HTTP)**
- Other possible levels:
 - **IP Level (Layer 3)** using tools like Iptables

OSI Model Layers

None

Layer 1 → Physical

Layer 2 → Data Link

Layer 3 → Network

Layer 4 → Transport

Layer 5 → Session

Layer 6 → Presentation

Layer 7 → Application

Avoid Being Rate Limited (Client-Side Best Practices)

- Use **client-side caching**
- Understand API limits
- Avoid sending too many requests in a short time
- Handle exceptions properly
- Add **retry with backoff strategy**

4. Key Takeaway

A complete rate limiter design includes:

- Choosing the right algorithm
- Handling distributed challenges
- Optimizing performance
- Monitoring and refining system behavior

Done with Rate Limiter Design 